

Automatic Room-Lock and Lights Circuit

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This circuit counts and indicates up to nine persons present in the room and automatically locks the door and switches off lights as soon as they leave. It automatically switches on the lights when the first person enters the room. The circuit can be used as a power-saving device and a security device to prevent unauthorised entry in the room, especially in a business meeting room.

Circuit and working

The circuit, shown in Fig. 1, is built around three NE555 timer ICs (IC1, IC2 and IC6), an up/down BCD decade counter 74LS192 (IC3), a 7-segment display driver 74LS47 (IC4), 4-input NOR gate HEF4002B (IC5) and a com-

mon-anode 7-segment display LTS542 (DIS1). S1 and S2 are foot switches. S1 is installed under door-mat of the door used to enter the room, and S2 is installed under a door-mat just inside the room.

The outputs of IC3 change state synchronously with the low-to-high transitions on the clock inputs. Separate count up (CPU) and count down (CPD) pins of IC3 are used here. Its parallel pin 11 is made high and parallel data input pins 15, 1, 10 and 9, along with master reset pin 14, are made low in this circuit.

The outputs (Q0 through Q3) of IC3 are given to IC4 to drive display DIS1. Resistor R11 is used to limit the current flowing through DIS1.

Outputs of IC3 are also given to IC5. Output O1 of IC5 is fed to relay-driver transistor T3 through resistor

Test Points	
Test point	Details
TP0	0V, GND
TP1	+5V
TP2	Low for around 5 seconds when switch S1 is pressed momentarily
TP3	Low for around 5 seconds when switch S2 is pressed momentarily
TP4	High and light is off. Display shows '0,' means room is vacant
TP5	High and door is locked. Display shows '0,' means room is vacant
TP6	High for around 11 seconds when switches S3 and S4 are momentarily pressed simultaneously. Also lock remains open for 11 seconds

R9 to energise relay RL1. Light switches off when RL1 is energised. The second output O2 is fed to relay-driver transistor T4 to energise relay RL2 for locking the door. RL1 and RL2 energise simultaneously as inputs of 4-input

NOR gate (IC5) are connected to both the drivers.

Output O2 of IC5 is also given to the base of transistor T7 whose collector is connected to the base of transistor T6. When O2 of IC5 goes high, it locks the door and makes the Vcc pin 8 of IC1 low to stop further counting.

To open the lock, press switches S3 and S4 (acting as security keys) simultaneously. The monostable circuit built around NE555 (IC6) triggers and its output goes high for around 11 seconds and

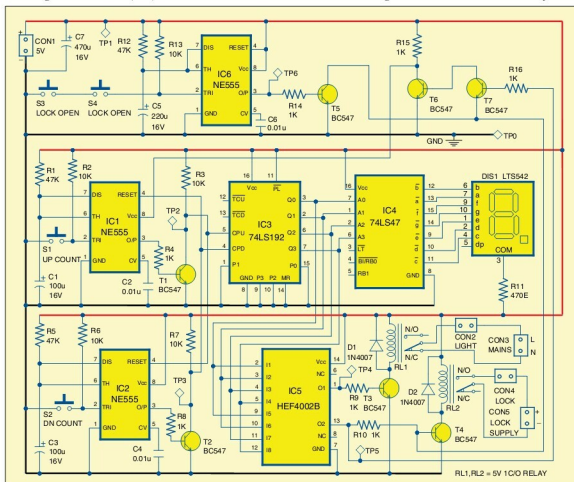


Fig. 1: Circuit diagram of the automatic lock and lights circuit